



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	AAI

Name of Student:	BARI ANKIT VENOD
Roll No. :	61
Assignment No.:	3
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	3
Demonstrated Knowledge	3	2
Legibility	2	2
Total	10	07

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Vaishali Shirsat

Signature :

Date :

17/13

2. i) Compare / contrast undercomplete and overcomplete autoencoders.

→ (i) Undercomplete autoencoders -

an autoencoder where:

$$\text{hidden layer size} < \text{input size}$$

- It compresses the input into a lower-dimensional representation.
- Key characteristics:
 - learns dimensionality reduction
 - forces model to capture most important features
 - similar to PCA
 - less risk of simply copying input.
- Advantages:
 - learns compact representation
 - good for feature extraction
 - works well without special regularization
- disadvantages:
 - may lose important information.

(ii) Overcomplete autoencoder -

an autoencoder where -

$$\text{hidden layer size} > \text{input size}$$

- Key characteristics:
 - has more neurons than input
 - can easily learn identity function
 - requires regularization
 - higher hidden neuron size
- Advantages:
 - can learn richer representation
 - useful for complex data
- disadvantages:
 - risk of learning trivial identity mapping.
 - requires constraints

Q. 2) Compare / contrast denoising and sparse autoencoders.

→ (i) denoising autoencoder (DAE)

- Idea - Add noise to input and train model to reconstruct clean version.

$$\tilde{x} = x + \text{noise}$$

- Train -

$$\text{Reconstruction loss} = ||x - \hat{x}||^2$$

- Purpose - learn robust features

- Advantages

- removes noise
- learn stable representations.
- improves generalization

(ii) Sparse autoencoder (SAE) -

- Idea - force most hidden neurons to be inactive.
- add sparsity constraints -

$$\sum KL(p || \hat{p}_j)$$

where, p = density sparsity ρ
 $\hat{p}$ = actual activation

- Purpose - learn meaningful feature detectors.

- Advantages

- Avoids trivial identity mapping
- learns interpretable features.
- denoising robust to corrupt
- but sparse forces selective neuron activation
- both prevent identity mapping in overcomplete AE.

3> Compare / contrast general autoencoders and variational autoencoders.

→ (i) General encoders - (vanilla)

- structure - encoder $\rightarrow$ bottleneck $\rightarrow$ decoder
- flows - $x \rightarrow z \rightarrow \hat{x}$
- which minimizes reconstruction loss only.
- characteristics - deterministic latent vector
 - No probabilistic modeling
 - cannot generate new realistic samples well.
- General AE - good for feature learning.

(ii) Variational autoencoders (VAE) -

- Introduced by, Diederik P. Kingma and Max Welling.
- Paper - auto-encoder variational Bayes.
- core idea - instead of learning a fixed vector z ,
 - learn a distribution,

$$z \sim N(\mu, \sigma^2)$$

- VAE loss function -

$$L = \text{Reconstruction Loss} + \text{KL Divergence}$$

$\therefore$ most specifically,

$$L = \|x - \hat{x}\|^2 + \text{KL}(q(z|x) || p(z))$$

- VAE - true generative model
- VAE latent space is smooth, can interpolate between points
- VAE allow sampling new realistic images.

Q. 4) Explain the architecture of variational autoencoders in details.

→ (i) Encoder network -

input = x

output $\rightarrow \mu(x), \sigma(x)$

↳ Not a single vector, two vectors.

(ii) Representation trick -

- instead of sampling directly:

$$z \sim \mathcal{N}(\mu, \sigma^2)$$

So,

$$z = \mu + \sigma \cdot \epsilon, \text{ epsilon } \mathcal{N}(0, 1)$$

why?

- makes sampling differentiable
- enable backpropagation

(iii) Decoder network -

- Takes sampled z and reconstructs

$$z \rightarrow \hat{x}$$

(iv) Loss function components -

(a) Reconstruction loss -

- measures, how close is output to input?
- Eg, MSE (continuous data)
- Binary cross entropy (images)

(b) KL divergence -

$$KL(q(z|x) || p(z))$$

forces latent distribution close to standard normal -

$$p(z) = \mathcal{N}(0, 1)$$